Estimating Effective Elastic Properties of 3D Printed Solid and Lattice Samples Using Resonant Ultrasound Spectroscopy

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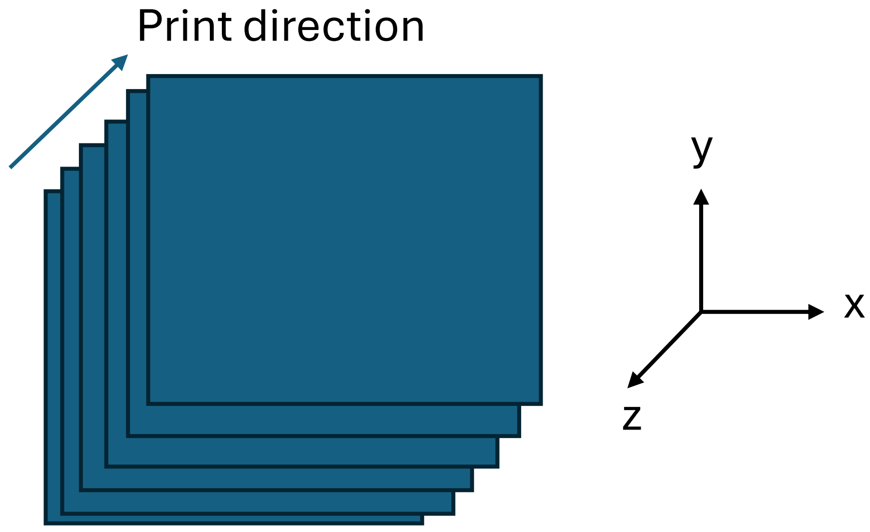
Material Anisotropy

Isotropic

$$\begin{bmatrix}\sigma_{11}\\ \sigma_{22}\\ \sigma_{33}\\ \sigma_{23}\\ \sigma_{13}\\ \sigma_{12}\end{bmatrix} = \begin{bmatrix} C_{11} & C_{12} & C_{12} & 0 & 0 & 0\\ C_{12} & C_{11} & C_{12} & 0 & 0 & 0\\ C_{12} & C_{12} & C_{11} & 0 & 0 & 0\\ 0 & 0 & 0 & C_{44} & 0 & 0\\ 0 & 0 & 0 & 0 & C_{44} & 0\\ 0 & 0 & 0 & 0 & 0 & C_{44}\end{bmatrix}\begin{bmatrix}\epsilon_{11}\\ \epsilon_{22}\\ \epsilon_{33}\\ 2\epsilon_{23}\\ 2\epsilon_{13}\\ 2\epsilon_{12}\end{bmatrix}$$

$$C_{11} = \lambda + 2\mu$$
$$C_{12} = \lambda$$
$$C_{44} = \mu$$

λ : Lame's first parameter
μ: Lame's second parameter



Layer by layer printing process introduce material anisotropy

Tetragonal

$$\begin{bmatrix}\sigma_{11}\\ \sigma_{22}\\ \sigma_{33}\\ \sigma_{23}\\ \sigma_{13}\\ \sigma_{12}\end{bmatrix} = \begin{bmatrix} C_{11} & C_{12} & C_{13} & 0 & 0 & 0\\ C_{12} & C_{11} & C_{13} & 0 & 0 & 0\\ C_{13} & C_{13} & C_{33} & 0 & 0 & 0\\ 0 & 0 & 0 & C_{44} & 0 & 0\\ 0 & 0 & 0 & 0 & C_{44} & 0\\ 0 & 0 & 0 & 0 & 0 & C_{66}\end{bmatrix}\begin{bmatrix}\epsilon_{11}\\ \epsilon_{22}\\ \epsilon_{33}\\ 2\epsilon_{23}\\ 2\epsilon_{13}\\ 2\epsilon_{12}\end{bmatrix}$$

Six independent elastic constants

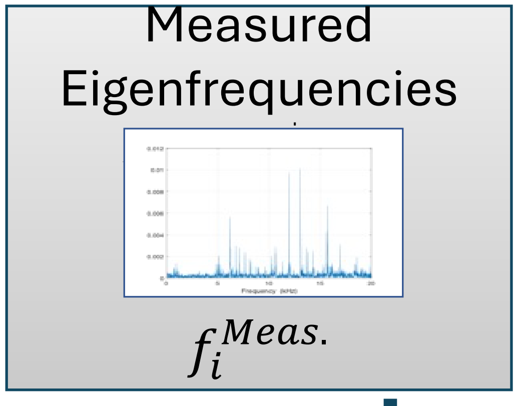
$$E_1 = (C_{33}C_{11}^2 + 2C_{13}^2\,C_{12} - 2C_{11}C_{13}^2 - C_{33}C_{12}^2)/(C_{11}C_{33} - C_{13}^2)$$

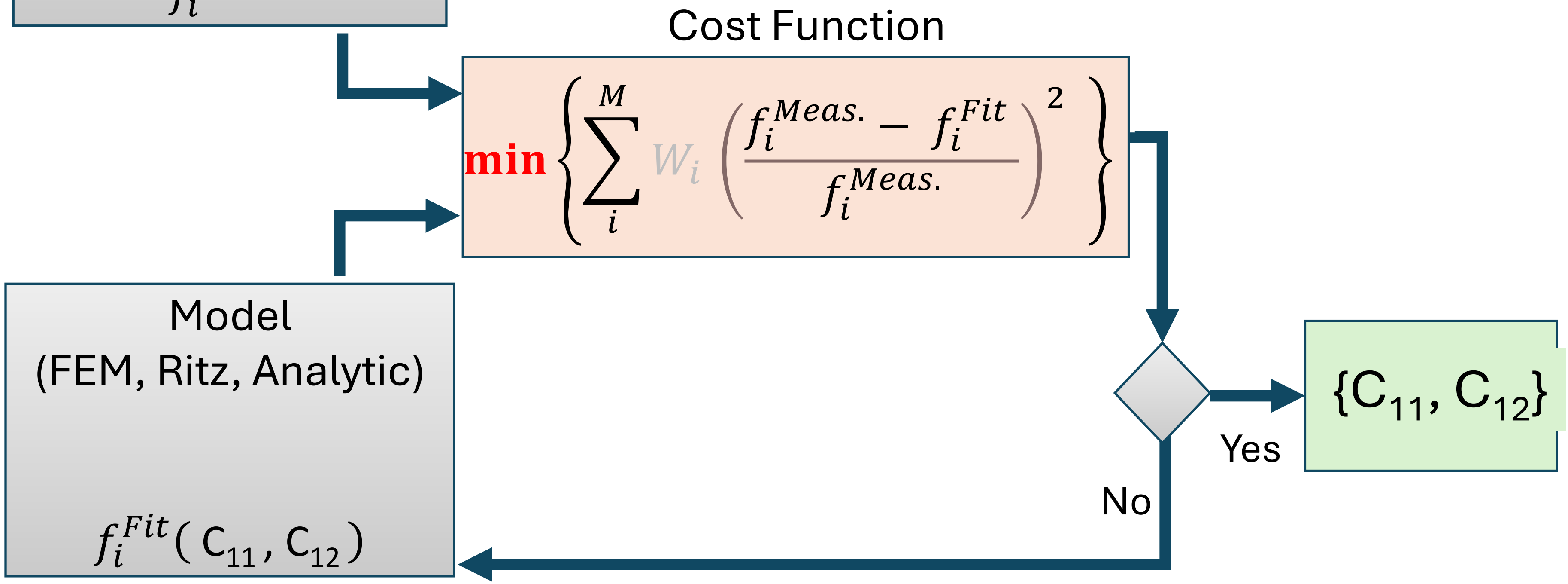
$$E_1 = E_2$$

$$E_3 = (C_{33}C_{11}^2 + 2C_{13}^2\,C_{12} - 2C_{11}C_{13}^2 - C_{33}C_{12}^2)/(C_{11}^2 - C_{12}^2)$$

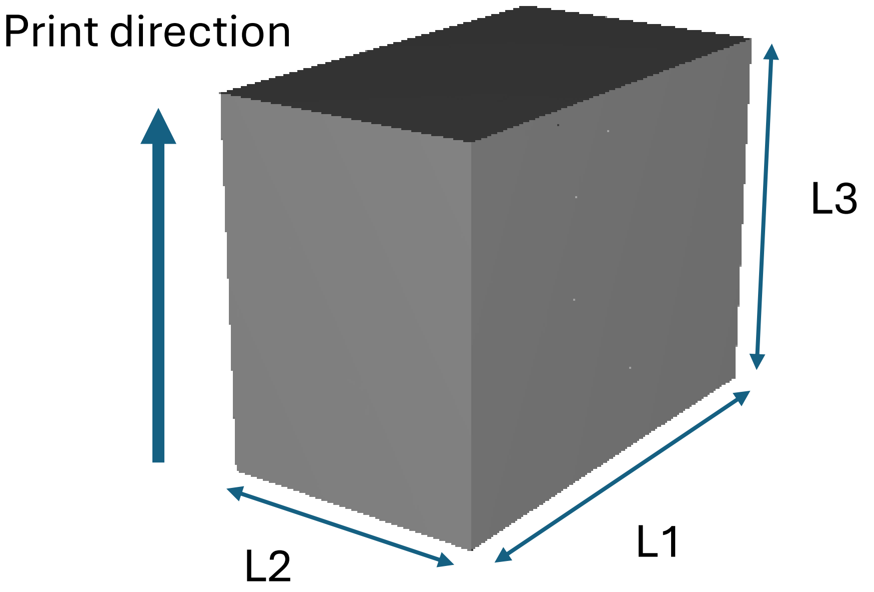
Estimating Elastic Coefficients

Resonant Ultrasound Spectroscopy relies on the ability to accurately estimate the **natural frequencies** of an elastic solid

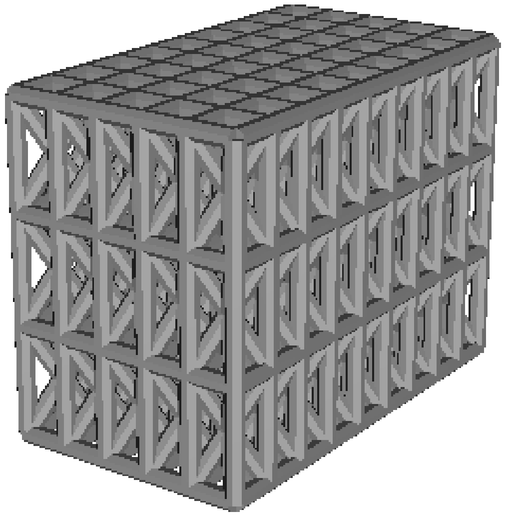




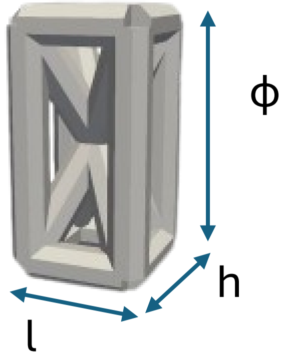
Methods: Sample Design



Solid Sample



Lattice Sample

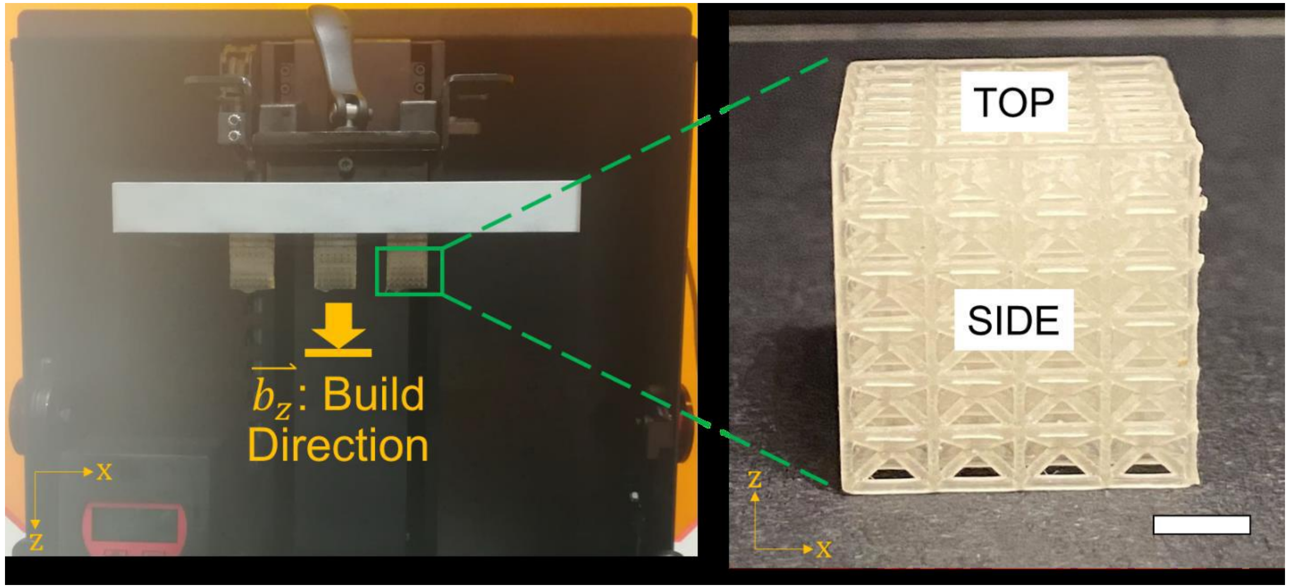


Unit Cell Dimension	l	h	ϕ
(mm)	2.20	5.00	0.80

Base Material	Density (kg/m^3)	Poisson Ratio	Tensile Modulus (Mpa)
Amber	1120	0.33	3000

Sample Dimension	L1	L2	L3
(mm)	20.60	11.80	15.80

Methods: Printing Process



Process: Stereolithography
Layer thickness : 50 μm
Layer exposure time: 5,000 ms

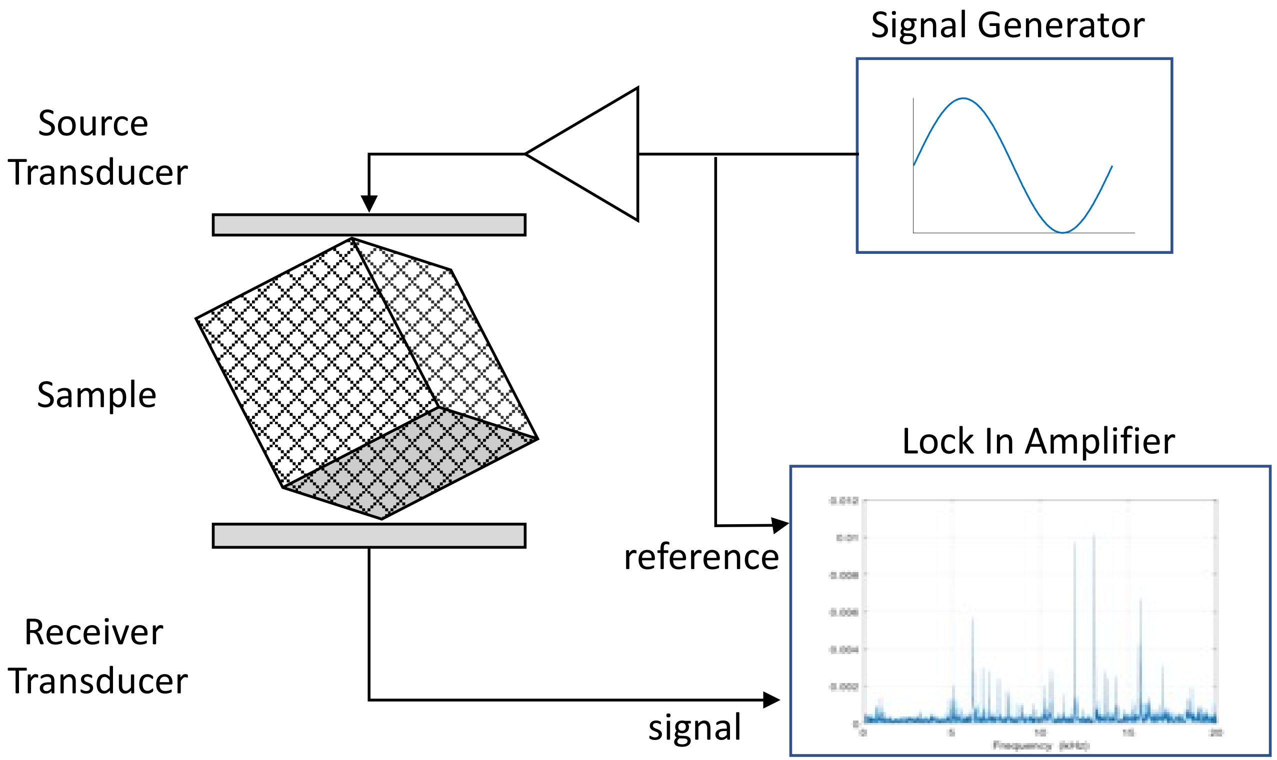
Solid Sample

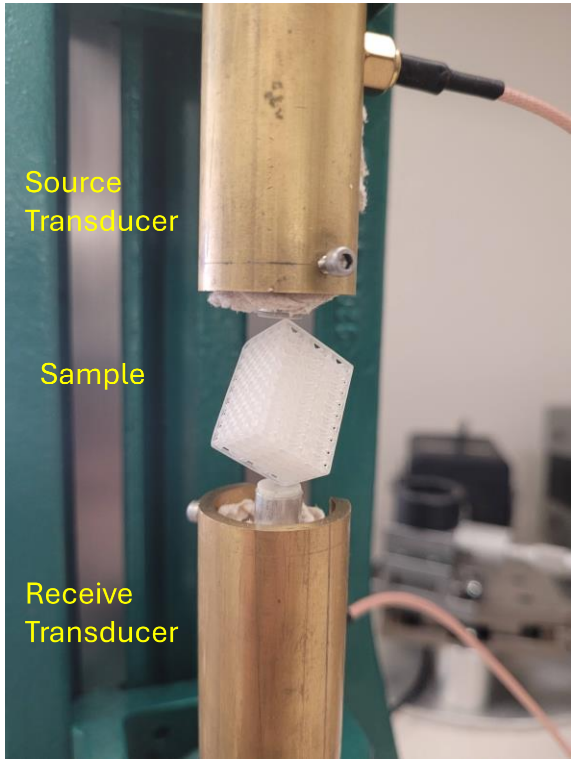
Parameters	Sample 1	Sample 2	Sample 3
L1 (mm)	20.072	19.851	19.871
L2 (mm)	10.896	11.241	11.261
L3 (mm)	15.345	14.748	14.76
mass (g)	4.074	3.998	3.994

Lattice Sample

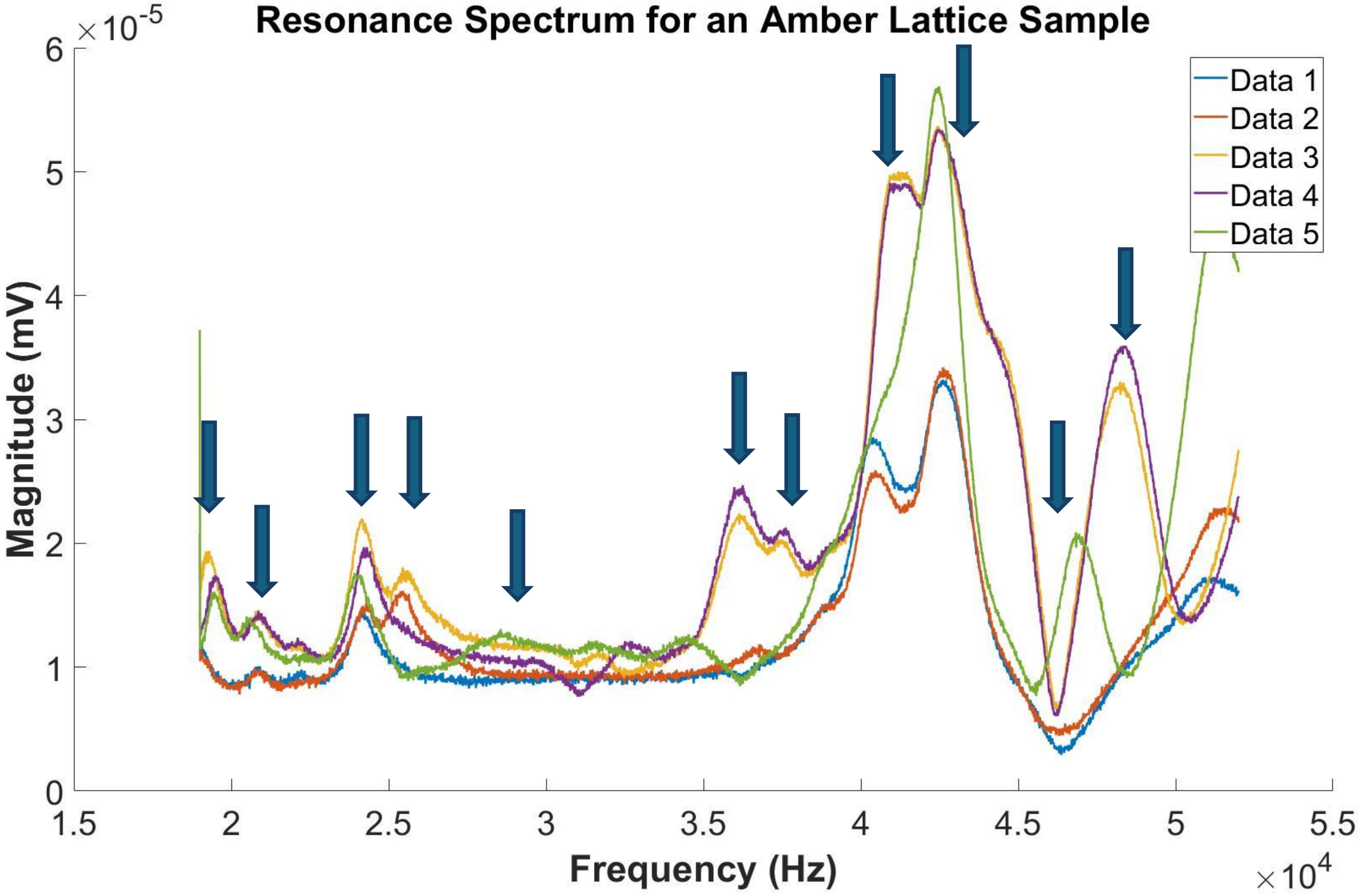
Parameters	FEA	Sample 2a	Sample 2b
L1 (mm)	20.6	20.7	20.7
L2 (mm)	11.78	11.81	11.81
L3 (mm)	15.79	15.87	15.87
mass (g)	2.29	2.26	2.26
Density (kg/m^3)	564	583	583

Methods: Resonant Ultrasound Spectroscopy Experiment





Eigen Frequency (Solid Sample)



Eigen Frequency (Solid samples)

Mode	Sample1	Sample2	Sample 3
1	20724	20765	20620
2	27980	27424	27708
3	30872	30355	30486
4	34186	33710	35636
5	37524	36946	38000
6	38394	38245	37998
7	39252	39640	39836
8	40068	39840	40164
9	40716	41030	41395
10	43100	43132	43244
11	44220	44875	Undetected
12	45804	45974	45406
13	46365	46160	46355
14	48632	49713	49128

Estimating Elastic Coefficients (Solid Sample)

Dimension
Density

Calculation order : 9

Resonance Frequency: 10

Symmetry: **Isotropic**

Iterations: 5

Lower bound:
0.001× Elasticity coefficients
Upper bound:
5× Elasticity coefficients

RUS Algorithm

Output

- Model Frequencies
- Engineering and Voight Coefficients

Elasticity coefficients: C11, C44 (for isotropic case)

Results

	E (Gpa)	C11	C44
Sample 1	2.97	5	1.09
Sample 2	3.12	5	1.16
Sample 3	3.04	5	1.12

Base material property: 3 GPa

Eigen Frequency (Lattice Sample)

- **Long Wavelength Regime**
  - Excitation Wavelength >> Smallest *Structural* length scale (strut length)
  - Assuming a clamped-clamped beam

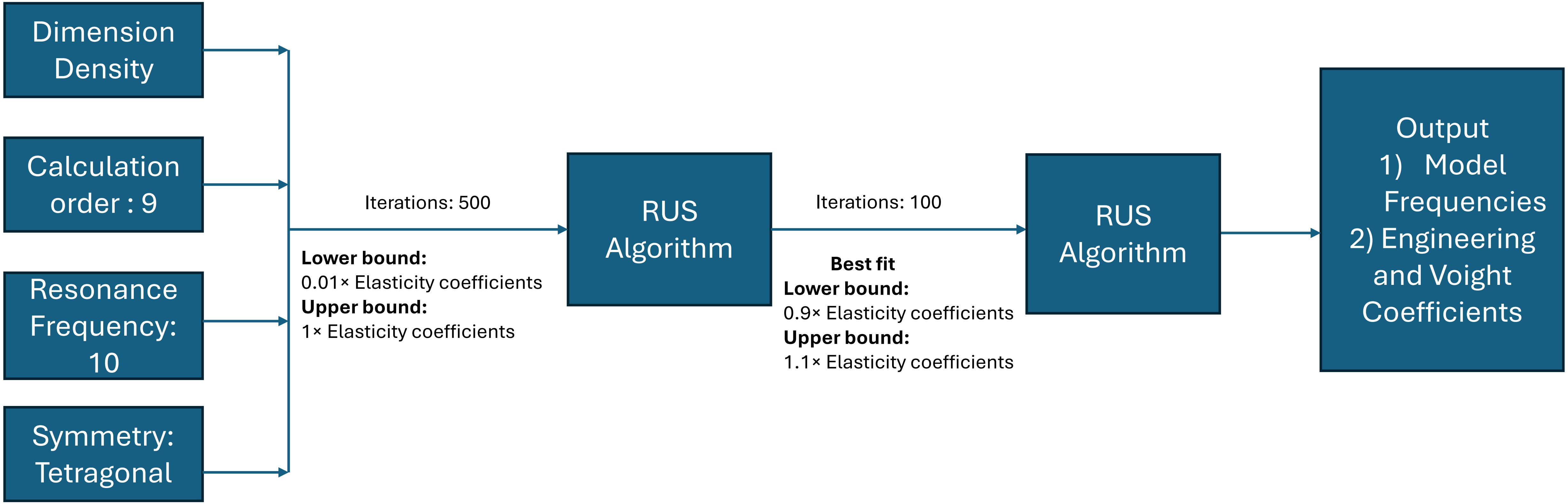
$$f_{RUS}^{max} < \left((1.506\pi)^2 \sqrt{\frac{E_o I}{\rho_o A L_s{}^4}}\right) \sim 33\ kHz$$

$$\left(\rho_o: 1120\,\frac{kg}{m^3},\ L_s \sim 5.90\text{ mm},\ D_s \sim 0.8\text{ mm},\ E_o: 3000\text{ MPa}\right)$$

Eigen Frequency (Lattice sample)

Mode	FEM	Sample 2a	Sample 2b
1	11286	8720	8300
2	13805	12230	11870
3	17306	Undetected	13730
4	19187	14750	14150
5	19800	Undetected	15950
6	21112	Undetected	16600
7	21359	16880	Undetected
8	21784	Undetected	17500
9	26207	19730	19130
10	28232	22820	22490

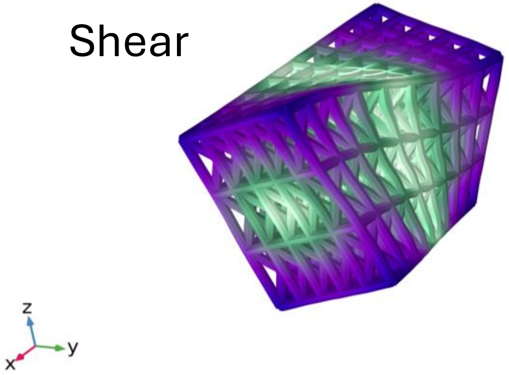
Estimating Elastic Coefficients (Lattice Sample)



Eigen Frequencies and Mode shapes of the Lattice Sample

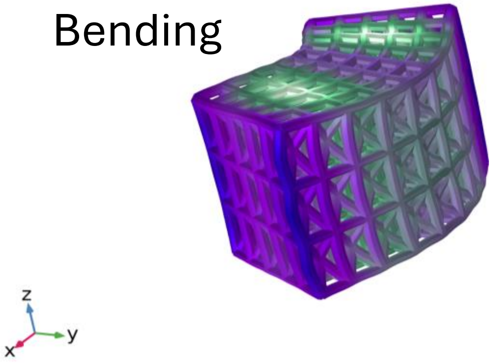
11286 Hz

Shear



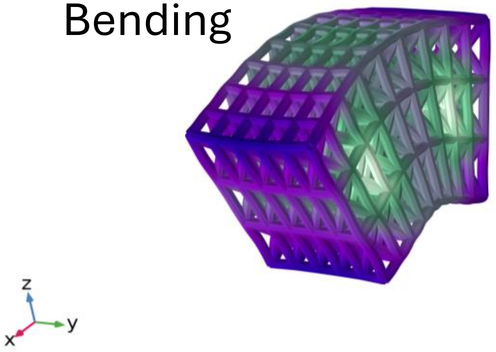
13805 Hz

Bending



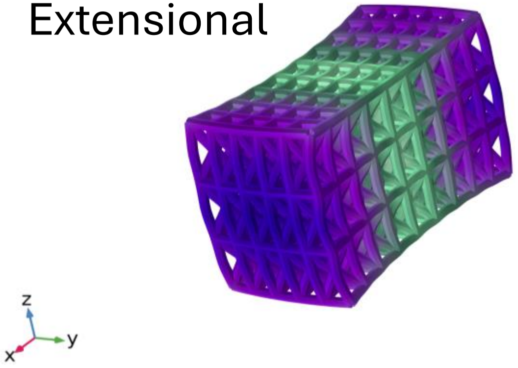
17306 Hz

Bending



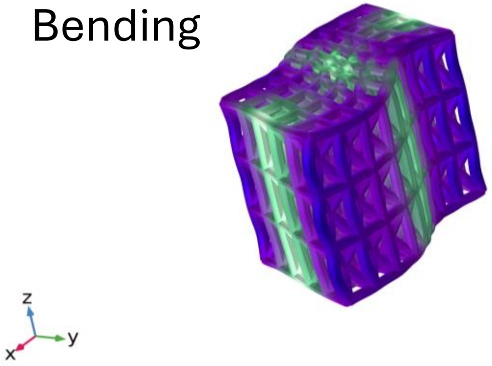
19187 Hz

Extensional



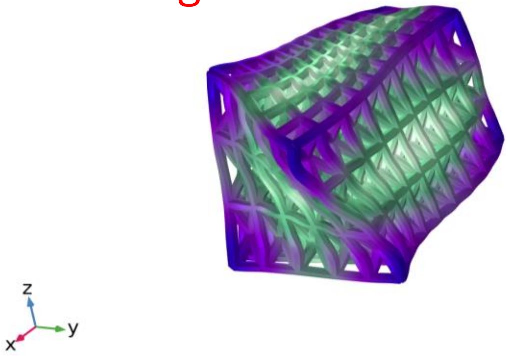
19800 Hz

Bending



21112 Hz

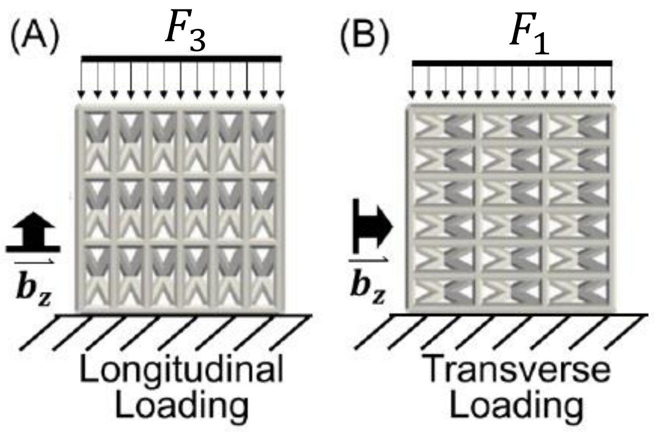
Bending + Shear



Elasticity Coefficients and estimated Moduli of the Lattice

	C_{11}	C_{12}	C_{13}	C_{33}	C_{44}	C_{66}	Fit (%)
Model (FEA)	0.58	0.27	0.13	0.95	0.22	0.11	0.006
Sample 2a	0.61	0.31	0.11	0.94	0.24	0.14	2.30
Sample 2b	0.52	0.31	0.11	0.76	0.18	0.09	4.52

	E_1 (Gpa)	E_2 (Gpa)	E_3 (Gpa)
Model (FEA)	0.451	0.451	0.907
Sample 2a	0.452	0.452	0.923
Sample 2b	0.34	0.334	0.732
Quasistatic simulation (effective)	0.244	0.244	0.661



Summary and Conclusions

- Amber lattice material is difficult to characterize due to losses of a number of detectable modes
- Sample size is approaching limit of continuum approximation (need larger sample)
- Differences in porosity in lattice samples has a significant impact on true value of elastic modulus of the lattice parts
- Differences between ‘*as built*’ vs ‘*as modeled*’ results suggest further work is needed to resolve and understand these issues.

References

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